```
set sql_notes = 0;
drop database if exists schoolmanagement;
set sql_notes = 1;

create database schoolmanagement;
use schoolmanagement;


create table Students(student_id int primary key auto_increment,
            first_name varchar(100) not null,
            last_name varchar(100) not null);

create table Subjects(subject_id int primary key auto_increment,
            subject_name varchar(100) not null unique);

create table Exams(exam_id int primary key auto_increment,
            exam_name varchar(100) not null unique,
            exam_date date not null);

create table Marks(mark_id int primary key auto_increment,
            student_id int not null,
            subject_id int not null,
            exam_id int not null,
            score int not null check(score between 0 and 100),
            foreign key(student_id) references students(student_id),
            foreign key(subject_id) references subjects(subject_id),
            foreign key(exam_id) references exams(exam_id));

-- Students
insert into students(first_name,last_name) values
('Aarav','Sharma'),
('Priya','Patel'),
('Rohan','Verma'),
('Ananya','Singh'),
('Karthik','Reddy'),
('Sneha','Iyer');

-- Subjects
insert into subjects(subject_name) values
('Mathematics'),
('Science'),
('Hindi'),
('English'),
('Social Studies');

-- Exams
insert into exams(exam_name,exam_date) values
('Unit Test 1 2024','2024-07-15'),
('Half-Yearly 2024','2024-10-10'),
('Annual 2025','2025-03-20');

-- Marks (student_id, subject_id, exam_id, score)
insert into marks(student_id,subject_id,exam_id,score) values
-- Unit Test 1 2024
(1,1,1,78),(1,2,1,85),(2,1,1,60),
-- Half-Yearly 2024
(1,1,2,92),(1,2,2,88),(1,3,2,76),
(2,1,2,81),(2,4,2,69),
(3,2,2,55),(4,3,2,90),(5,5,2,72),
-- Annual 2025
(1,4,3,84),(2,2,3,73),(3,1,3,48),(4,5,3,95),(6,3,3,67);

#Q1. Retrieve all students: first name and last name of every student
```

```
select first_name, last_name from students;

#Q2. Filter by score: all entries in Marks where score is greater than 80
select * from marks where score > 80;

#Q3.  Sort data: all exams sorted by date, most recent first
select * from exams order by exam_date desc;

#Q4.  Student marks: first name, subject name and score for every mark
select s.first_name, sub.subject_name, m.score
from marks m
join students s on m.student_id = s.student_id
join subjects sub on m.subject_id = sub.subject_id;

-- Q5. Subject averages: subject name and average score
select sub.subject_name, avg(m.score) as average_score
from marks m
join subjects sub on m.subject_id = sub.subject_id
group by sub.subject_id, sub.subject_name;

-- Q6. Count exams: how many marks are recorded for "Aarav Sharma"
select count(*) as total_marks
from marks m
join students s on m.student_id = s.student_id
where s.first_name = 'Aarav' and s.last_name = 'Sharma';



-- Q7. Top performer: student with the highest single score in any exam
select first_name, last_name
from students
where student_id in (select student_id from marks
                     where score = (select max(score) from marks));

-- Q8. Scores by exam: exam name and total score, only where total > 250
select e.exam_name, sum(m.score) as total_score
from marks m
join exams e on m.exam_id = e.exam_id
group by e.exam_id, e.exam_name
having sum(m.score) > 250;

-- Q9. Pass/Fail label: student name, subject, score and status (Pass if 70 or above)
select concat(s.first_name,' ',s.last_name) as student_name,
       sub.subject_name,
       m.score,
       case when m.score >= 70 then 'Pass' else 'Fail' end as status
from marks m
join students s on m.student_id = s.student_id
join subjects sub on m.subject_id = sub.subject_id;
```